§ 11-1-012

甜菊醣苷
Steviol glycosides

別　　名：

熱水萃取法製得　：INS No. 960a

酵素修飾法製得　：尚無

定　　義：本品來自甜菊(*Stevia rebaudiana* Bertoni)葉片，以甜菊醇 (steviol) 為骨幹，並與任意數量或組合之醣類(葡萄糖、鼠李糖、木糖、果糖、阿拉伯糖、半乳糖和去氧葡萄糖)形成以共價鍵與醣基結合之結構。

熱水萃取法製得　：本品以熱水萃取甜菊葉片，萃取液以樹脂吸附並濃縮其中甜菊醣苷。以乙醇溶劑清洗脫附，獲得甜菊醣苷粗產品。粗產品再以甲醇或乙醇水溶液重新結晶，亦可使用離子交換樹脂純化。終產品可經由噴霧乾燥獲得。

酵素修飾法製得　：本品以經基因轉殖之非產毒及非病原性微生物 (*Pichia pastoris* 及 *Escherichia coli*)生產之酵素(葡萄糖基轉移酶 glucosyltransferase 及蔗糖合成酶 sucrose synthase)處理由甜菊葉片萃取純化之甜菊糖苷，再經加熱使酵素失去活性並過濾去除，獲得酵素修飾之甜菊醣苷粗產品。粗產品經樹脂吸附/脫附或固/液相過濾等濃縮步驟，再經由脫色、結晶化及噴霧乾燥進行純化製得。

酵素生產使用之基因轉殖微生物:

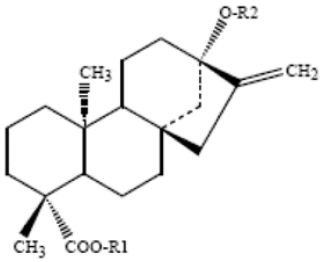
1. *Pichia pastoris* (基因來源包含 *Horedum vulgare* L、*Stevia rebaudiana* Bertoni、*Vigna radiate*)。
2. *Escherichia coli* (基因來源包含 *Acidithiobacillus caldus*、*Arapidopsis thaliana*、*Solanum tuberosum*、*Stevia rebaudiana* Bertoni)。

化學名稱　　：另列於《附加規定》

C.A.S.編號　：另列於《附加規定》

化學式　：另列於《附加規定》

結構式　：



甜菊醇（R1 = R2 = H）為甜菊醣苷之糖苷配基。

Glc、Rha、Fru、deoxyGlc、Gal、Ara 及 Xyl 依序代表葡萄糖 (glucose)、鼠李糖 (rhamnose)、果糖 (fructose)、去氧葡萄糖 (deoxyglucose)、半乳糖 (galactose)、阿拉伯糖 (arabinose) 及木糖 (xylose)。

含量　：總含量在 95%以上。以乾重計。

外　　觀　：白至淡黃色粉末，無臭或輕微特殊氣味。甜度約為蔗糖之 200 ~ 300 倍。

特　　性

鑑　　別

溶解度　：極微溶到易溶於水；微溶到易溶於乙醇水溶液(50:50,v/v)

HPLC 層析圖形　：與標準品相符

pH 值　：4.5 ~ 7.0 (1%溶液)

純　　度

灰分　：1%以下

乾燥減重　：6%以下 (105℃，2 小時)

殘留溶劑　：甲醇在 200 mg/kg 以下

乙醇在 5000 mg/kg 以下

砷　：1 mg/kg 以下

鉛　：1 mg/kg 以下

微生物規範　：總生菌數：1000 CFU/g 以下

酵母菌及黴菌：200 CFU/g 以下

大腸桿菌：陰性/1g

沙門氏桿菌：陰性/25g

分　　類：食品添加物第（十一之一）類。

用　　途：甜味劑。

《附加規定》

<附加規定>

Summary of Formula, Molecular weight, steviol equivalent and sugar moieties in Identified Steviol Glycosides from the Leaves of *Stevia rebaudiana* Bertoni. [Adapted from Purkayastha & Kwok (2020)]

#	Common Name	CAS Number	Trivial Formula	Mol. Wt	Steviol Equivalent	R1	R2	Reference
1. Steviol + Glucose (SvGn)								
1.01	Steviolmonoside		SvG1	481	0.66	H	Glcβ1-	Ohta et al. (2010)
1.02	Steviolmonoside A		SvG1	481	0.66	Glcβ1-	H	Gardana et al. (2010)
1.03	Rubusoside	64849-39-4	SvG2	643	0.49	Glcβ1-	Glcβ1-	Ohta et al. (2010)
1.04	Steviolbioside	41093-60-1	SvG2	643	0.49	H	Glcβ(1–2)Glcβ1-	Kohda et al. (1976)
1.05	Stevioside	57817-89-7	SvG3	805	0.40	Glcβ1-	Glcβ(1–2)Glcβ1-	Bridel and Lavielle (1931)
1.06	Stevioside A		SvG3	805	0.40	Glcβ(1–2)Glcβ1-	Glcβ1-	Wu et al. (2012)
1.07	Rebaudioside B	58543-17-2	SvG3	805	0.4	H	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Kohda et al. (1976)
1.08	Rebaudioside G		SvG3	805	0.4	Glcβ1-	Glcβ(1–3)Glcβ1-	Ohta et al. (2010)
1.09	Stevioside B		SvG3	805	0.4	Glcβ(1–3)Glcβ1-	Glcβ1-	Chaturvedula and Zamora (2014)
1.10	Rebaudioside E	63279-14-1	SvG4	967	0.33	Glcβ(1–2)Glcβ1-	Glcβ(1–2)Glcβ1-	Sakamoto et al. (1977a)
1.11	Rebaudioside A	58543-16-1	SvG4	967	0.33	Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Kohda et al. (1976)
1.12	Rebaudioside A2		SvG4	967	0.33	Glcβ1-	Glcβ(1–6)Glcβ(1–2)Glcβ1-	Chaturvedula and Prakash (2011a)
1.13	Rebaudioside D	63279-13-0	SvG5	1129	0.28	Glcβ(1–2)Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Sakamoto et al. (1977a)
1.14	Rebaudioside I		SvG5	1129	0.28	Glcβ(1–3)Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)
1.15	Rebaudioside L		SvG5	1129	0.28	Glcβ1-	Glcβ(1–6)Glcβ(1–2)[Glcβ(1–3)] Glcβ1-	Ohta et al. (2010)
1.16	Rebaudioside Q2		SvG5	1129	0.28	Glcα(1–2)Glcα(1–4) Glcβ1-	Glcβ(1–2)Glcβ1-	Chaturvedula and Prakash (2011b)
1.17	Rebaudioside Q		SvG5	1129	0.28	Glcβ1-	Glcα(1–4)Glcβ(1–2)[Glcβ(1–3)]	-

<附加規定>

Summary of Formula, Molecular weight, steviol equivalent and sugar moieties in Identified Steviol Glycosides from the Leaves of *Stevia rebaudiana* Bertoni. [Adapted from Purkayastha & Kwok (2020)]

						Glcβ1-		
1.18	Rebaudioside I2		SvG5	1129	0.28	Glcβ1-	Glcα(1–3)Glcβ(1–2)[Glcβ(1–3)] Glcβ1-	Chaturvedula et al. (2011a)
1.19	Rebaudioside Q3		SvG5	1129	0.28	Glcβ1-	Glcα(1–4)Glcβ(1–3)[Glcβ(1–2)] Glcβ1-	Chaturvedula et al. (2011a)
1.20	Rebaudioside I3		SvG5	1129	0.28	Glcβ(1–2)[Glcβ(1–6)] Glcβ1-	Glcβ(1–2)Glcβ1-	Chaturvedula et al. (2011a)
1.21	Rebaudioside AM	2222580-26-7	SvG5	1129	0.28	Glcβ(1–2)[Glcβ (1–3)] Glcβ1-	Glcβ(1–2)Glcβ1-	Prakash and Ma (2018)
1.22	Rebaudioside M	1220616-44-3	SvG6	1291	0.25	Glcβ(1–2)[Glcβ (1–3)] Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)
1.23	Rebaudioside 1h		SvG7	1453	0.22	Glcβ(1–3)Glcβ(1–2)[Glcβ (1–3)]Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Prakash and Ma (2018)
1.24	Rebaudioside IX		SvG9	1778	0.18	Glcβ(1–2)[Glcβ(1–3)] Glcβ1-	Glcβ(1–3){Glcβ(1–3) [Glcβ(1–2)] Glcα (1–6)Glcβ(1–2)} Glcβ1-	Prakash and Ma (2018)
2. Steviol + Rhamnose + Glucose (SvR1Gn)								
2.01	Dulcoside A	64432-06-0	SvR1G2	789	0.40	Glcβ1-	Rhaα(1–2)Glcβ1-	Kobayashi et al. (1977)
2.02	Dulcoside B		SvR1G2	789	0.40	H	Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)
2.03	Rebaudioside C	63550-99-2	SvR1G3	951	0.33	Glcβ1-	Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Sakamoto et al. (1977b)
2.04	Rebaudioside C2		SvR1G3	951	0.33	Rhaα(1–2)Glcβ1-	Glcβ(1–2)Glcβ1-	Purkayastha et al. (2019)
2.05	Rebaudioside S		SvR1G3	951	0.33	Rhaα(1–2)Glcβ1-	Glcα (1–2)Glcβ1-	Ibrahim et al. (2016)
2.06	Rebaudioside H		SvR1G4	1113	0.29	Glcβ1-	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)] Glcβ1-	Ohta et al. (2010)
2.07	Rebaudioside K		SvR1G4	1113	0.29	Glcβ(1–2)Glcβ1-	Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)
2.08	Rebaudioside K2		SvR1G4	1113	0.29	Glcβ(1–6)Glcβ1-	Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha et al. (2019)
2.09	Rebaudioside J		SvR1G4	1113	0.29	Rhaα(1–2)Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)
2.10	Rebaudioside N	1220616-	SvR1G5	1275	0.25	Rhaα(1–2)[Glcβ(1–	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)

〈附加規定〉

Summary of Formula, Molecular weight, steviol equivalent and sugar moieties in Identified Steviol Glycosides from the Leaves of *Stevia rebaudiana* Bertoni. [Adapted from Purkayastha & Kwok (2020)]

		46-5				3)] Glcβ1-		
2.11	Rebaudioside N2		SvR1G5	1275	0.25	Glcβ(1–2)[Glcβ (1–3)] Glcβ1-	Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Prakash and Ma (2018)
2.12	Rebaudioside N6		SvR1G5	1275	0.25	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Glcβ(1–2)Glcβ1-	Prakash and Ma (2018)
2.13	Rebaudioside O	1220616-48-7	SvR1G6	1437	0.22	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Ohta et al. (2010)
2.14	Rebaudioside O2		SvR1G6	1437	0.22	Glcβ(1–4)Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha (2016)
2.15	Rebaudioside O5		SvR1G6	1437	0.22	Glcβ(1–2)[Glcβ (1–3)] Glcβ1-	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)] Glcβ1-	Prakash and Ma (2018)
2.16	Rebaudioside O6		SvR1G7	1600	0.20	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Glcβ(1–6)Glcβ(1–3)[Glcβ(1–2)] Glcβ1-	Prakash and Ma (2018)
2.17	Rebaudioside O7		SvR2G6	1584	0.20	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)]Glcβ1-	Glcβ(1–3)Rhaα(1–2)[Glcβ(1–3)] Glcβ1-	Prakash and Ma (2018)
3. Steviol + Xylose + Glucose (SvX1Gn)								
3.01	Stevioside F		SvX1G2	775	0.41	Glcβ1-	Xylβ(1–2)Glcβ1-	Chaturvedula and Prakash (2011c)
3.02	Rebaudioside F	438045-89-7	SvX1G3	937	0.34	Glcβ1-	Xylβ(1–2)[Glcβ(1–3)]Glcβ1	Starratt et al. (2002)
3.03	Rebaudioside F2		SvX1G3	937	0.34	Glcβ1-	Glcβ(1–2)[Xylβ(1–3)]Glcβ1-	Chaturvedula and Prakash (2011c)
3.04	Rebaudioside F3		SvX1G3	937	0.34	Xylβ(1–6)Glcβ1-	Glcβ(1–2)Glcβ1-	Chaturvedula et al. (2011b)
3.05	Rebaudioside R		SvX1G3	937	0.34	Glcβ1-	Glcβ(1–2)[Glcβ(1–3)] Xylβ1-	Ibrahim et al. (2016)
3.06	Rebaudioside U		SvX1G4	1099	0.29	Xylβ(1–2)Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha et al. (2019)
3.07	Rebaudioside U2		SvX1G4	1099	0.29	Xylβ(1–2)[Glcβ(1–3)] Glcβ1-	Glcβ(1–2)Glcβ1-	Purkayastha (2016)

<附加規定>

Summary of Formula, Molecular weight, steviol equivalent and sugar moieties in Identified Steviol Glycosides from the Leaves of *Stevia rebaudiana* Bertoni. [Adapted from Purkayastha & Kwok (2020)]

3.08	Rebaudioside U3		SvX1G4	1099	0.29	Xylβ(1–2)[Glcβ(1–4)] Glcβ1-	Glcβ(1–2)Glcβ1-	Purkayastha et al. (2019)
3.09	Rebaudioside V		SvX1G5	1261	0.25	Glcβ(1–2)[Glcβ(1–3)] Glcβ1-	Glcβ(1–2)[Xylβ(1–3)]Glcβ1-	Purkayastha et al. (2019)
3.10	Rebaudioside V2		SvX1G5	1261	0.25	Xylβ (1–2)[Glcβ(1–3)] Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Prakash and Chaturvedula (2013)
4. Steviol + Arabinose + Glucose (SvA1Gn)								
4.01	Rebaudioside W		SvA1G4	1098	0.29	Glcβ(1–2)[Araβ(1–3*)] Glcβ1	Glcβ(1–2)Glcβ1-	Purkayastha (2016)
4.02	Rebaudioside W2		SvA1G4	1098	0.29	Araβ(1–2*)Glcβ1	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha (2016)
4.03	Rebaudioside W3		SvA1G4	1098	0.29	Araβ(1–6)Glcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha et al. (2019)
4.04	Rebaudioside Y		SvA1G5	1260	0.25	Glcβ(1–2)[Araβ(1–3*)] Glcβ1	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha et al. (2019)
5. Steviol + Fructose + Glucose (SvF1Gn)								
5.01	Rebaudioside A3		SvF1G3	967	0.33	Glcβ1-	Glcβ(1–2)[Fruβ(1–3)]Glcβ1-	Chaturvedula et al. (2011c)
6. Steviol + Galactose + Glucose (SvGa1Gn)								
6.01	Rebaudioside T		SvGa1G4	1129	0.28	Glcβ(1–2)Glcβ1-	Galβ(1–2)[Glcβ(1–3)]Glcβ1-	Purkayastha (2016)
7. Steviol + Deoxyglucose + Glucose (SvdG1Gn)								
7.01	Stevioside D		SvdG1G2	789	0.40	Glcβ1-	6-deoxyGlcβ(1–2)Glcβ1-	Chaturvedula and Prakash (2011d)
7.02	Stevioside E		SvdG1G3	951	0.33	Glcβ1-	6-deoxyGlcβ(1–2)[Glcβ(1–3)] Glcβ1-	Chaturvedula and Prakash (2011d)
7.03	Stevioside E2		SvdG1G3	951	0.33	6-deoxyGlcβ1-	Glcβ(1–2)[Glcβ(1–3)]Glcβ1-	Chaturvedula et al. (2011d)